europass



Francesco Santoro

Data di nascita: 05/01/1994 | **Luogo di nascita:** Brescia, Italia | **Nazionalità:** Italiana |

Numero di telefono: (+39) 3200108001 (Cellulare) | **Indirizzo e-mail:** francesco.santoro.geologo@gmail.com | **Sito web:** https://fsdv.eu/ | **LinkedIn:** https://www.linkedin.com/in/francesco-santoro-geo/ |

Indirizzo: Viale Italia, 11, 25064, Gussago, Italia (Abitazione)

PRESENTAZIONE

I'm a geologist with a passion for applied mineralogy and building materials - from cultural heritage preservation to modern construction innovations and future space applications.

I've been part as ESR3 in the SUBLime project, a Marie Skłodowska-Curie Actions initiative at the University of Granada, where I focused on developing self-cleaning lime-based materials through biomimetic approaches, with particular studies on amorphous-crystalline transitions and surface modifications through both chemical and physical methods.

Currently, I'm a Research Fellow at the Center of Studies and Activities for Space "Giuseppe Colombo" (CISAS) at the University of Padova, working within the "Space It Up project" - a major initiative funded by the Italian Space Agency (ASI) and the Italian Ministry of University and Research (MUR). My research focuses on the characterization of terrestrial analogues and lunar regolith simulants for extraterrestrial construction, working on optimizing new mix formulations.

My research interests combine geological expertise with innovative approaches in materials science, using earth-based simulants to bridge the gap between terrestrial and space construction materials.

PATENTE DI GUIDA

Patente di guida: B

ESPERIENZA LAVORATIVA

TRAINERSHIP AND ERASMUS+ – PROF. CARLOS RODRIGUEZ NAVARRO – 31/01/2019 – 29/09/2019 – GRANADA, SPAGNA

Through the Erasmus+ scholarship, I actively participated in an interdisciplinary research group working closely with Post-Doc Luis Monasterio Guillot. During the internship and development of my master's thesis, I improved my problem-solving and analytical skills by working from raw samples to finished products through data modelling. I utilized the following techniques:

- Single crystal diffraction and powder diffraction (XRD and XRD2D)
- Vibrational spectroscopy (Raman and infrared techniques)
- Chemical and textural investigation techniques (X-ray fluorescence, SEM in WDS, EDS, TEM)
- Mechanical strength tests and geotechnical tests

SERVIZIO CIVILE, AMBITO AMBIENTALE – UFFICIO TECNICO DEL COMUNE DI GUSSAGO – 14/12/2019 – 14/04/2020 – GUSSAGO, ITALIA

The project was interrupted due to the Covid-19 pandemic. This experience provided me with exposure to public administration and land management, encompassing both regulatory compliance and environmental support, with direct involvement in geological reporting and contaminated site remediation

MARIE SKŁODOWSKA-CURIE ACTION EUROPEAN TRAINING NETWORK – INNOVATIVE TRAINING NETWORK (ETN ITN) ESR 3. – UNIVERSITÀ DEGLI STUDI DI GRANADA (UGR) – 04/2021 – 15/04/2024 – GRANADA, SPAGNA

Impresa o settore: Istruzione | **Sito Internet:** https://sublime-etn.eu/

I've been part as ESR3 in the SUBLime project, a Marie Skłodowska-Curie Actions initiative at the University of Granada, where I focused on developing self-cleaning lime-based materials through biomimetic approaches, with particular studies on amorphous-crystalline transitions and surface modifications through both chemical and physical methods. The title of the project is "Lime-based mortars and plasters with added functionality: a biomimetic approach".

RESEARCHER ASSISTANT CONTRACT IN UGR – UNIVERSITY OF GRANADA (UGR) – 05/2024 – 11/2024 – GRANADA, SPAGNA

Research focused on surface modification techniques and chemical etching processes formaterial characterization and enhancement.

RESEARCH FELLOW – CENTER OF STUDIES AND ACTIVITIES FOR SPACE "GIUSEPPE COLOMBO" (CISAS) – 15/01/2025 – 15/01/2027 – PADOVA, ITALIA

WP9.3.3: terrestrial analogs analyses

ISTRUZIONE E FORMAZIONE

30/09/2017 – 24/09/2020 Padova, Italia
LAUREA MAGISTRALE IN GEOLOGIA E GEOLOGIA TECNICA Università degli studi di Padova

Relatore: Prof. Luca Valentini
Co-Relatore: Prof. Carlos Rodriguez Navarro
Co-Relatore: PhD Louis Monasterio

Indirizzo Via Gradenigo, 6, 35131, Padova, Italia | **Campo di studio** Mineralogia Applicata, Cristallochimica dei materiali |

Voto finale 105/110 | **Tipo di crediti** 123 CFU | **Tesi** "Process of carbonation and durability in alternative binders"

30/09/2014 – 16/07/2017 Padova, Italia
LAUREA TRIENNALE IN SCIENZE GEOLOGICHE Università degli studi di Padova

Relatore: prof. Richard Spiess

Indirizzo Via Gradenigo, 6, 35131, Padova, Italia | **Campo di studio** Petrologia | **Tipo di crediti** 180 CFU |

Tesi "Understanding the granulitic deformation microstructures of Curinga metagabbro: A SEM-EBSD study"

15/04/2021 – 09/10/2026 Granada, Spagna
PHD Universidad de Granada

Tesi: "LIME-BASED MORTARS AND PLASTERS WITH ADDED FUNCTIONALITY: A BIOMIMETIC APPROACH"

REFERENZE

Carlos Rodríguez Navarro Full Professor (UGR)

Head of the Mineralogy Scientific Department of Granada and Doctoral Advisor

E-mail carlosrn@ugr.es | **Telefono** (+34) 958246616

Encarnación Ruiz Agudo Full Professor (UGR)

Dean of the School of Sciences at UGR and Professor in the Mineralogy Scientific Department of Granada and PhD thesis supervisor

E-mail encaruiz@ugr.es | **Telefono** (+34) 958242366

Luca Valentini Associate Professor (UniPD)

Associate Professor in Applied Mineralogy at the University of Padova

E-mail luca.valentini@unipd.it | **Telefono** (+39) 0498279831

Matteo Massironi Associate Professor (UniPD)

Associate Professor in Planetary Geology at hte University of Paodva

E-mail matteo.massironi@unipd.it | **Telefono** (+39) 3484990807

ONORIFICENZE E RICONOSCIMENTI

09/2020

Finalist on European University Competition on Raw Materials, Mining engineering sector, 2020 – Raidmap Project

This international competition provided the opportunity to work in an advanced research environment alongside engineers, mathematicians, and physicists from all continents, addressing real-world industrial challenges. Focused on the mining engineering sector, this project enhanced both my communication skills and my ability to operate effectively in a high-level interdisciplinary setting.

05/2017

Finalist Padova University Mobility Challenge – Unismart - E-ON Energy

In this competition, my team and I developed and presented a project aimed at improving urban mobility in a medium-sized city like Padua. Our project advanced to the final stages of the competition organized by the University of Padua and E-ON Energy.

CONFERENZE E SEMINARI

06/2019 Ronda (Spagna)

XXXVIII Convegno interazionale della società geologica spagnola di Mineralogia

Moderator and member of the organizing staff

12/05/2019 Granada -UGR- (Spagna)

Magic of geopolymers: Alkaline-activated cement between past, present and future

Keynote speaker in the UGR department

Link http://grados.ugr.es/geologia/pages/geocervezas

08/2018 Padova (Italia)

Workshop sulla protezione del patrimonio culturale dal rischio sismico

Sino- Italian Workshop, CIPAR

17/05/2021 – 18/05/2021 Online

Kick-off meeting event SUBLime

10/05/2021 – 14/05/2021 Granada

PHREEQC course

Course delivered by Full Professor Encarnacion Ruiz Agudo, expert PhReeQc user

19/07/2021 – 23/07/2021 Granada

Micro-Raman course

Micro-Raman spectroscopy course delivered by the manufacturer HORIBA

12/07/2021 – 16/07/2021 Granada

Corso di microscopia a scanzione elettronica (SEM)

SEM training course conducted by specialized Thermo Fisher technicians, with particular focus on environmental scanning electron microscopy applications (without metallic coating)

13/09/2021 – 16/09/2021 Unversidad de Granada

Total Scattering Workshop

Total Scattering Workshop led by Prof. Jorge A. Rodriguez-Navarro and expert faculty from the University of Insubria, covering total scattering techniques and software applications including DEBUSSY for data analysis.

22/11/2021 – 26/11/2021 Universidad de Granada

School on properties of lime, SUBLime RTC1-1

Course delivered by Prof. Carlos Rodriguez Navarro within the SUBLime framework, extensively covering lime properties with emphasis on future research applications from both historical conservation and innovative materials perspectives.

10/07/2022 – 15/07/2022 Honolulu, Hawai'i, USA

Goldschmidt 2022 – International Conference of geochemistry in Honolulu

Goldschmidt International Conference, Honolulu, Hawai'i, USA. Delivered oral presentation at this premier international geochemistry conference.

30/11/2022 – 02/12/2022 Granada

9th Granada-Münster Discussion Meeting

Discussion Meeting with the Granada and Munster group with Helmut Coelfen, Andrew Putnis e Carlos rodriguez Navarro.

25/04/2022 – 29/04/2022 Santorini, Greece

Workshop in Geochemical Modelling, SUBLime RTC 1-3

Full Geochemical Modelling couse with Prof. Encarnación Ruiz-Agudo.

19/09/2022 – 23/09/2022 Darmstadt, Germany

Workshop on the sustainability of lime, SUBLime RTC1-4

Network Research Training on the Sustainability of lime, in the contest of RTC1-4

24/02/2023 – 25/02/2023 Cordoba, Spain (Feb 24-25)

Marie Curie Alumni Association Annual Conference

06/03/2023 – 11/03/2023 Rabat, Morocco (Mar 6-11)

RILEM – Spring Convention CMSS23

11/04/2023 – 12/04/2023 Online

SUBLime – Scientific Progress conference meet- ing

13/06/2023 – 15/06/2023 Ghent, Belgium

CrysPom VIII – Crystallization in Porous Media conference

09/07/2023 – 14/07/2023 Lyon, France

Goldschmidt 2023 – International Conference of geochemistry in Lyon

22/05/2023 – 26/05/2023 TU Delft, Nederland

Workshop on Micro-Meso Scale Structures, SUBLime RTC 2-1

Micro-Meso Scale Structures RTC 2-1, held by Prof. Erik Schlangen in TU Delft, Nederland

23/10/2023 – 27/10/2023 UMinho, Portugal

Workshop on Lime Mortars Masonry system, Sublime RTC1-2

Workshop on Lime Mortars Masonry system, in the frame of Sublime RTC1-2 with Prof. Paulo B. Lourenço

18/10/2023 – 21/10/2023 Guimarães, Portugal

Marie Curie Soft Skills and Comunication skills training

19/04/2024 – 27/04/2024 Gliwice, Poland

Workshop on Structure design, SUBLime RTC 2-2

Structure design, RTC 2-2, SUBLime Network Training

17/04/2024 – 19/04/2024 Gliwice, Poland

Complementary skills Workshop in the SUBLime Network

Complementary skills Workshop
SUBLime Network Training
Gliwice, Poland (Apr 17-19)

26/08/2024 – 30/08/2024 Padova, Italy
ECM34 – 34th European Crystallographic Meeting

ECM34 – 34th European Crystallographic Meeting Padova, Italy

30/08/2024 – 02/09/2024 Padova, Italy
EPDIC18 – 18th European Powder Diffraction Conference

EPDIC18 – 18th European Powder Diffraction
Conference

09/11/2024 – 14/11/2024 Madeira, Portugal
SUBLime – international conference

27/11/2024 – 30/11/2024 Granada Spain
11th Granada-Münster Discussion Meeting

10/02/2025 – 14/02/2025 hybrid
2025 Europlanet GMAP Winter School

The GMAP Winter School is focused on the production of planetary geological maps of a range of Solar System bodies: building upon earlier editions.

Link https://www.planetarymapping.eu/

27/01/2025 – 30/01/2025 Padova
Chemometrics course of Experimental Design

"Chemometrics: Experimental Design" course, instructed by Prof. Riccardo Leardi from the University of Genoa, is scheduled to take place from January 27 to 30, 2025. The course duration is 30 hours in total (5 CFU).

14/04/2025 – 17/04/2025 Metz, France
International Workshop: Granular Materials Fundamentals, Applications and Concepts for Extraterrestrial Regolith including Moon, Mars and Asteroids

Link https://regolith-metz.event.univ-lorraine.fr/

20/05/2025 Padova
Lithium in igneous and volcano-sedimentary systems: geological aspects and exploration strategies

Link https://www.geoscienze.unipd.it/en/lithium-igneous-and-volcano-sedimentary-systems-geological-aspects-and-exploration-strategies-0

22/06/2025 – 27/06/2025 Munster, Germany
European Lunar Symposium

Link https://cris.uni-muenster.de/portal/de/project/132729696

06/07/2025 – 11/07/2025 Prague, Czech Republic
Goldschmidt 2025 – International Conference of geochemistry in Prague

Link https://conf.goldschmidt.info/goldschmidt/2025/goldschmidt/2025/meetingapp.cgi

02/09/2025 – 04/09/2025 Padova
7th Historic Mortars Conference (HMC 2025)

Actively participating in the organization and scientific production, HMC 2025 is a conference covering the technology and characterization of historic mortars, issues related to the conservation and restoration of historic masonry structures and archaeological sites, the development of heritage-inspired building materials for sustainable

architecture, the formulation of materials for architectural restoration and test methods for historic and restoration mortars.

Link https://hmc2025.com/

16/09/2025 – 18/09/2025 Padova
Le Geoscienze e le sfide del XXI secolo, Congresso congiunto SIMP-SGI Padova

Conference featuring two presentations on the work carried out in the first year of the Space It Up project

Link https://www.geoscienze.org/padova2025/

26/01/2026 – 28/01/2026 Firenze
Space it Up! Days

The event represents the first plenary congress of the Space it Up! project, bringing together project partners and the broader scientific community. Three poster presentation.

Link https://spaceitup.it/event/siu-days/

20/03/2026 Padova
Convegno Albo Cavatori del Veneto: Presentazione progetto LIFE Svolta Blu

Link https://www.festari.it/convegno-albo-cavatori-del-veneto-presentazione-progetto-life-svolta-blu/

21/06/2026 – 26/06/2026 Nancy
European Lunar Symposium 2026

The European Lunar Symposium (ELS) is an annual international conference that brings together researchers, engineers, and professionals working on **lunar science and exploration providing** a key forum for presenting **cutting-edge research, fostering collaborations, and discussing Europe's role in the future of lunar exploration**. Here I presented a work on micro tomography on alkali activated materials for lunar construction.

02/07/2026 – 03/07/2026 Milano
Mining forward: exploration, innovation and society

Gain a multidisciplinary perspective on global and national mineral exploration trends, exploring advanced geophysical methods, innovative technologies for the recovery of critical raw materials (CRM) from mining waste, and the socio-regulatory impact of activities on the local environment.

Link https://www.unimi.it/it/corsi/corsi-post-laurea-e-formazione-continua/corsi-di-dottorato-phd/catalogo-degli-insegnamenti-la-formazione-dottorale/aa-2025/2026-mining-forward-exploration-innovation-and-society

08/07/2026 – 10/07/2026 Aachen & online
GTT-Technologies Users' Meeting 2026

How to use and learn FactSage, ChemApp, ChemSheet

02/09/2026 – 04/09/2026 Alba
Space It Up! Young Minds Summer Forum

The **Space It Up! Young Minds Summer Forum** is a three-day event designed for **early-career researchers** working on space-related topics comining the learning experience of a summer school with the scientific exchange of a research forum.
Participants will share their work, exchange perspectives across disciplines, and contribute to shaping the future of Space It Up! and the wider space ecosystem.

Link https://spaceitup.it/event/space-it-up-phd-forum/

01/08/2026 – 09/08/2026 Firenze
COSPAR 2026

06/09/2026 – 11/09/2026 Nairobi
80th RILEM Week 2026 in conjunction with the 3rd ACCTA International Conference on Advances in Climate-Friendly Construction Technology in Africa

The 80th RILEM Week will be held in conjunction with the 3rd RILEM Conference on Advances in Climate-Friendly Construction Technology in Africa (ACCTA) in **Nairobi, Kenya**, from **5 to 11 September 2026**. The conference will be the first one ever **in the very pulse of Africa**.

Link https://rilemnairobi26.sciencesconf.org/

12/09/2026 – 20/09/2026 Ries

PANGAEA ESA project 2026

ESA's PANGAEA training course prepares astronauts and space engineers to identify planetary geological features for future missions to the Moon, Mars and asteroids.
The PANGAEA campaign – named after the ancient supercontinent – provides the crew with introductory and practical knowledge to find interesting rock samples as well as to assess the most likely places to find traces of life on other planets. Leading European planetary geologists share their insights into the geology of the Solar System.
Course participants of the seventh edition in 2026 are ESA astronaut Pablo Álvarez Fernández, NASA astronaut Jasmin Moghbeli and Kimiya Yui from the Japanese space agency, JAXA.
The theory part is followed by field trips to the Italian Dolomites, the Ries crater in Germany and the volcanic landscapes of Lanzarote in Spain. I partecipate for the Logistic and like geological expert.

Link https://www.esa.int/Science_Exploration/Human_and_Robotic_Exploration/CAVES_and_Pangaea/What_is_PANGAEA | https://blogs.esa.int/caves/category/pangaea/

16/11/2026 – 17/11/2026 Roma

Workshop "Space It Up!: Un motore per l'innovazione scientifica e tecnologica"

A due anni dall'avvio delle attività, il Workshop "Space It Up!: Un motore per l'innovazione scientifica e tecnologica" rappresenta l'occasione per **valorizzare i risultati più innovativi e maturi** raggiunti nell'ambito del progetto e per condividere con il mondo della ricerca, delle istituzioni e dell'industria nuove soluzioni, nuove opportunità e nuove prospettive per il futuro spaziale del Paese.

Link https://www.asi.it/event/workshop-space-it-up-un-motore-per-linnovazione-scientifica-e-tecnologica/

25/11/2026 – 27/11/2026 Granada

13th Granada-Münster Discussion Meeting

The Granada-Münster Discussion Meeting (GMDM) is an opportunity for researchers studying fluid-mineral interactions at all scales, from nanoscale to large-scale processes. In this occasion I will present an "exotic" presentation on geochemistry without water.

Link https://sites.google.com/go.ugr.es/gmdm/gmdm-13th?authuser=0

28/10/2026 – 16/12/2026 Padova

Corso spin-off Unipd "Dalla ricerca all'impresa: Percorso di imprenditorialità accademica"

Primo corso assoluto dell'Università di Padova per acquisire strumenti e competenze per valorizzare i risultati della propria ricerca, valutarne il potenziale di mercato e sviluppare un progetto imprenditoriale spin-off.
È organizzato dall'Ufficio Valorizzazione delle conoscenze e imprenditorialità accademica in collaborazione con Start Cube.

Link https://www.unipd.it/avviare-spin-off

COMPETENZE LINGUISTICHE

Lingua madre: **ITALIAN**

Altre lingue:

	COMPRENSIONE		ESPRESSIONE ORALE		SCRITTURA
	Ascolto	Lettura	Produzione orale	Interazione orale	
ESPAÑOL	C2	C1	C1	C1	C1
ENGLISH	C1	C1	B2	C1	C1
ΕΛΛΗΝΙΚΆ	A2	A1	A1	A2	A1

	COMPRENSIONE		ESPRESSIONE ORALE		SCRITTURA
	Ascolto	Lettura	Produzione orale	Interazione orale	
FRANÇAIS	A2	A1	A2	A2	A1

Livelli: A1 e A2: Livello elementare B1 e B2: Livello intermedio C1 e C2: Livello avanzato

COMPETENZE

Origin Pro | XPert High Score + | LaTEX | XPowderX | ImageJ | Posta elettronica | conoscenza di Windows e di Linux | Applicazioni grafiche e di publishing (Photoshop GIMP Illustrator InkScape Scribus) | Libre Office (Libre Office Writer, Libre Office Impress, Libre Office Calc)

HOBBY E INTERESSI

Music

From playing guitar and saxophone to listening to jazz, blues, and bossa nova.

Photography, visual arts, decorative arts, and cultural heritage

From landscape and architectural photography to visiting art galleries, from studying decorative techniques to exploring cultural sites.

Link https://www.juzaphoto.com/me.php?p=232357&pg=allphotos&srt=data&l=it

Sport

From football to surfing, from table tennis to sumo.

PUBBLICAZIONI

2024
Additive impact on early-stage magnesium carbonate mineralization.

Santoro De Vico F., Elert K., Rodríguez-Navarro C., Bonilla-Correa S., Pelayo-Punzano G., Ruiz-Agudo E. Additive impact on early-stage magnesium carbonate mineralization. In: Geochemical Perspectives Letters, 32 (2024), 1–6

2025
Formation of micropillars on etched calcite and their biomimetic effect on wettability: Implications for built heritage protection

Santoro De Vico, F., Elert, K., Ruiz-Agudo, E., & Rodriguez-Navarro, C. (2025). Formation of micropillars on etched calcite and their biomimetic effect on wettability: Implications for built heritage protection. Materials & Design, 255.

2023
A Physico- chemical Study of Lime-based Mortars from Different Historical Periods

Santoro De Vico F., Zagaroli A., Rodríguez-Navarro C., Kubica J.,Gorski M. A Physico- chemical Study of Lime-based Mortars from Different Historical Periods. In: NanoWorld Journal, 9S2 (Sept. 2023), S256–S261

2025
Mudbricks with tea grounds: A zero CO2 emission option for more sustainable and environmentally-friendly construction

Crespo-López, Laura, et al. "Mudbricks with tea grounds: A zero CO2 emission option for more sustainable and environmentally-friendly construction." Construction and Building Materials 472 (2025): 140982.

2025
Multi-Analytical Assessment of Ethyl Silicate-Treated Bricks: From Salt Deterioration to Long-Term Protection

Crespo-López, Laura, et al. "Multi-Analytical Assessment of Ethyl Silicate-Treated Bricks: From Salt Deterioration to Long-Term Protection." Journal of Building Engineering (2025): 113384.

2024

Novel Techniques for the Assessment of Injectability & Environmental Resistance of Injection Grouts for Mosaic Conservation

Nome della pubblicazione: Proceedings of the Workshop “Lime Based Materials for Repairing Historic Structures”

2026

olcanic deposits from mount Etna (Italy) as high-fidelity lunar simulants for In-Situ Resource Utilization (ISRU) applications

Autori: Giacomo Melchiori, Francesco Santoro De Vico, Alice Dottori, Riccardo Pozzobon, Luca Valentini, Patrizia Ferretti, Alessandro Bonetto, Michéle Lavagna, Sonia Calvari, Matteo Massironi | **Nome della pubblicazione**: Materials Today Advances | **Volume, numero, pagine**: Volume 29, 2026, 100678, ISSN 2590-0498

2026

Chemical-mineralogical features and physical properties of archaeological adobe: The evidence from Tell Zurghul/Nigin (Dhi Qar, Iraq)

Scrivi qui la descrizione

Chemical-mineralogical features and physical properties of archaeological adobe: The evidence from Tell Zurghul/Nigin (Dhi Qar, Iraq) L Volpi, F Santoro De Vico, A Arizzi, N Lanzaro, D Nadali Plos one 21 (2), e0342009

Autori: Luca Volpi, Francesco Santoro De Vico, Anna Arizzi, Nicola Lanzaro, Davide Nadali | **Nome della pubblicazione**: Plos one | **Volume, numero, pagine**: 21 (2), e0342009 | **Editore**: Public Library of Science

2026

Insights into the mechanisms of REE removal by structurally analogous gypsum and brushite phases

Insights into the mechanisms of REE removal by structurally analogous gypsum and brushite phases R Millán-Becerro, MP Asta, MB Ruíz, S Bonilla-Correa, FS De Vico, … Chemical Geology, 123304

Autori: Ricardo Millán-Becerro, María P Asta, Miguel Burgos Ruíz, Sarah Bonilla-Correa, Francesco Santoro De Vico, Encarnación Ruiz-Agudo | **Nome della pubblicazione**: Chemical Geology | **Volume, numero, pagine**: 123304 | **Editore**: Elsevier

2026

Coupling Seismic, Gravity and Magnetic Surveys for the Investigation of Planetary Lava Tubes: A Terrestrial Analog Case Study in Lanzarote (Spain)

Coupling seismic, gravity and magnetic surveys for the investigation of planetary lava tubes: A terrestrial analog case study in Lanzarote (Spain). Journal of Geophysical Research: Planets, 131, e2026JE009802

Autori: A. Ghirotto, I. Barone, F. Santoro De Vico, G. Melchiori, A. Zunino, E. Armadillo, A. Mittelholz, F. Sauro, M. Massironi | **Nome della pubblicazione**: Journal of Geophysical Research: Planets | **Volume, numero, pagine**: Volume 131, Issue 8 | **Editore**: AGU

2026

Role of amorphous phase, temperature fluctuations, and vacuum, on the performance of alkali-activated lunar regolith simulants

Role of amorphous phase, temperature fluctuations, and vacuum, on the performance of alkali-activated lunar regolith simulants

Autori: Anas Driouich · Ahmad Goudarzi · Francesco Santoro · Maria Chiara Dalconi · Massimo Dau · Carlo Bettanini · Shima Pilehvar · Luca Valentini | **Nome della pubblicazione**: Materials and Structures | **Volume, numero, pagine**: Volume 59, article number 349 (2026) | **Editore**: Springer Nature

RESEARCH EXPERIENCES

01/06/2023 – 01/09/2023

PhD secondment in UGent

University of Ghent (UGENT), Ghent, Belgium. Development of mortars with different functionalities.

10/07/2023 – 25/07/2023
Research Period µ-CT Core Facility (UGCT), Ghent, Belgium

Conducted extensive research on microtomography.

01/06/2022 – 01/07/2022
ARQUS Research Period University of Padova, Padova

Focused on development rheology studies

01/02/2022 – 15/03/2022
Industrial secondment Lhoist Group R & D, Nivelles, Belgium

Development of new mix design for the industry

TEACHING EXPERIENCE

2022 – 2024
Lecturer of "Techniques for Analyzing Geomaterials" in UGR

Two years of teaching experience (1 ECTS credit per year) in the Geology degree program.
Course taught in Spanish.
– Course entitled: "Techniques for Analyzing Geomaterials"

02/2025 – 05/2025
Guest lecturer in the course of MINERAL PROCESSES AND ENVIRONMENTAL APPLICATIONS in UNIPD 2025

Guest lecturer about the study and applications of lime and the use of PHREEQC software

07/11/2025 – 07/11/2025
Guest lecturer in the course of PLANETARY GEOLOGY in UNIPD 2025

Guest lecturer about the study and applications of ISRU technique on the moon

31/08/2025 – 31/12/2025
Teaching assistance for the course "PRINCIPI DI GEOSCIENZE" in UNIPD 2025

13/09/2026 – 18/09/2026
PANGEA Training assistance

Addestramento a Ries ai Astronauti con Harald e Carolyn van der Bogert

05/09/2026 – 07/09/2026
Invite lecture in the Pre-Conference Courses Circular strategies in sustainable construction.

During the The 80th RILEM Week will be held in conjunction with the 3rd RILEM Conference on Advances in Climate-Friendly Construction Technology in Africa (ACCTA) in **Nairobi, Kenya**, from **5 to 11 September 2026**. During the conferences I teach Circular strategies in sustainable construction.

01/10/2026 – 31/12/2026
Teaching assistance for the course "PRINCIPI DI GEOSCIENZE" in UNIPD 2026

RETI E AFFILIAZIONI

2021 – ATTUALE
RILEM

Link https://www.rilem.net/

2022 – ATTUALE
Geochemical Society

Link https://www.geochemsoc.org/

2021 – ATTUALE
SUBLime Network

2024 – ATTUALE
Giovani Cristallografi italiani

2025 – ATTUALE
Consorzio "SPACE IT UP"

2021 – ATTUALE
MCAA - Marie Curie Alumni association

Link https://www.mariecuriealumni.eu/

2017 – ATTUALE
University of Padua Alumni Association

Link https://www.unipd.it/en/alumni

2025 – ATTUALE
Europlanet

Link https://www.europlanet.org/

2025 – ATTUALE
SIMP - Società Italiana di Mineralogia e Petrologia

Link https://www.socminpet.it/

05/2025 – ATTUALE italia
Società Speleologica Italiana (SSI)

01/07/2026
AIDAA

COMPETENZE DI GESTIONE E DIRETTIVE

2010 – 2011
Student Representative, Liceo Calini, Brescia

Elected with the highest vote count as student representative. This experience allowed me to understand the dynamics within the school while actively collaborating and communicating with school administration and teachers.

2010 – 2012
Student Council Representative, Liceo Calini, Brescia

Elected with the highest vote count to represent students on the Provincial Student Council. This experience provided opportunities to collaborate with representatives from other provincial schools and advocate for my high school at provincial assemblies. Served as co-organizer of the Brescia Provincial Art Day (GDA) for two years.

2016 – 2020
Student Representative, Department of Geosciences, University of Padova

Elected with the second-highest vote count to represent students in the Department of Geosciences. During this two-year tenure, I addressed faculty challenges and served as a bridge between professors and fellow students.

01/01/2025 – ATTUALE
Co-supervisor of two Master thesis student

During my academic career, I co-supervised two excellent students:

- Alberto Coradazzi, master's thesis entitled "Anorthosites and troctolites from the Flakstadøy Massif as Analogues of the Lunar Highlands and the Related Magnesium Series"
- Elodie Goyaux, master's thesis entitled "Mixture Design for a New Generation of Alkalinically Activated Materials by Lunar Simulants: From Grain Size to Activator Optimization"

ATTIVITÀ SOCIALI E POLITICHE

27/09/2025 – 28/09/2025 Padova
Science4All 2025

Science communication and outreach at "Science4All," an event organized by the University of Padua for the European Researchers' Night. Participated at Stand 46 on behalf of the CISAS department.

Link https://science4all.it/

Padova - piattaforma social del dipartimento di Geoscienze
Video interview per l'International Day of Women and Girls in Science

Interview for the celebration of women in research.

Link https://youtu.be/i-BiWLAqh1c?si=qXh7J_8BhJ3WQ35E

12/2025 Padova - piattaforma social del dipartimento di Geoscienze
Video Communication on Etnean simulants article

Interviewed for science outreach on the publication "Volcanic deposits from Mount Etna (Italy) as high-fidelity lunar simulants for In-Situ Resource Utilization (ISRU) applications." Produced in extended English version and short Italian version.

Link https://youtu.be/orKbDZ7eTls?si=9SyJ4PlKoe9gPTwx

Padova
Science Outreach and Interview "Detective Antropocene"

Delivered presentations and activities at high schools for the "Anthropocene Detective" project, which I conceived and developed. Event held as part of "Geosciences Friday," a public engagement initiative by the Department of Geosciences for Sustainable Development.

06/03/2026 Padova
Video presentation on students as landscape detectives with Google Earth

What is meant by the Anthropocene? And how can we use Google Earth to visualize how the landscape is evolving over time due to human activities, especially mining? This is the experience of the students at the Copernico Pasoli High School in Verona who recently participated in one of our "Geoscience Days." Using a number of reference locations, the students played detective, investigating the changes occurring in different parts of the world, from coasts to major urban centers.

Link https://www.youtube.com/shorts/L9MoiFEpWKU?app=desktop

ATTUALE Web
Blog on my website

Because geology is just the starting point. Each category has its own blog. Enter in the website, pick one and explore.

Link https://fsdv.eu/hobbies.html